

Question

A certain short-period element **X** is often used in nuclear power reactors. A certain halide of it reacts fully in N_2O_4 to obtain a white crystal **A**.

Now, thermogravimetric analysis is performed on **A**. It is known that there are two weight loss peaks. First, it loses 58.04% of its weight to obtain a white substance **B**. Continuing heating, it loses another 8.52% of its weight to obtain a substance **C** with a special structure. **C** contains 4 C_3 axes, all **X** chemical environments are the same, and the coordination number is 4.

Coincidentally, people have also discovered substances with structures similar to **C** in many transition metals. In a similar particle **M** composed of a certain metal from the fourth period, the ligand contains acetate, and the mass fraction of the metal is 26.88%. All metal atoms chemical environments are the same, and all are 6-coordinated. It is known that this particle **M** can be prepared in aqueous solution.

The following statements are correct:

- A.** All other options are incorrect
- B.** **X** atom's ground state contains two unpaired electrons.
- C.** The number of atoms of **A** is greater than **C**.
- D.** The mass fraction of oxygen in **M** is 39.5%
- E.** The mass fraction of oxygen in **M** is 44.1%
- F.** The symmetry of **M** is greater than **C**
- G.** **M** is a neutral particle

Answer

Correct Answer: **E**

Detailed Explanation

Short-period element **X** is often used in nuclear power reactors. In nuclear reactors, Be is often used as a neutron moderator and reflector, and B is often used as a neutron absorber. Therefore, **X** may be B, Be.

CHECKPOINT

1 PTS

X may be B, Be

The halide of **X** reacts with N_2O_4 , generally yielding anhydrous nitrate and an adduct of N_2O_4 ; considering that the molecular weight of the loss in the first step is very large, it is very likely that the added N_2O_4 is lost.

CHECKPOINT

1 PTS

Halide reacts with N_2O_4 , generally yielding anhydrous nitrate and an adduct of N_2O_4

CHECKPOINT

1 PTS

The added N_2O_4 is lost in the first step of weight loss

If **X** is B, **A** can be written as $B(NO_3)_3 \cdot xN_2O_4$. First of all, nitrates of B are not common; secondly, no matter what value x takes, it cannot correspond to this weight loss peak. Therefore, **X** is not B.

CHECKPOINT

1 PTS

Based on the weight loss ratio, it is determined that **X** is not B

If **X** is Be, **A** can be written as $Be(NO_3)_2 \cdot xN_2O_4$. At this time, when x is 2, the first step weight loss ratio is $184/317 = 58.0\%$, which meets the requirements of the question.

Therefore, x is 2, and the chemical formula of **A** is $Be(NO_3)_2 \cdot 2N_2O_4$. **X** is Be, option B is incorrect.

CHECKPOINT

1 PTS

The chemical formula of **A** is $Be(NO_3)_2 \cdot 2N_2O_4$

C contains 4 C_3 axes, all **X** chemical environments are the same, and the coordination number is 4. First, it is judged that the compound has a tetrahedral structure; therefore, it is highly probable that it contains four Be. It is easy to think of the classic structure of Be compounds $Be_4O(NO_3)_6$. According to the weight loss peak, it can be judged that the chemical formula is correct. Therefore, **C** is $Be_4O(NO_3)_6$. According to the chemical formula, option C is incorrect.

CHECKPOINT

1 PTS

C has a tetrahedral structure; contains four Be

CHECKPOINT

1 PTS

C is $\text{Be}_4\text{O}(\text{NO}_3)_6$

According to the question, the particles formed by **M** should have a similar metal-oxygen cluster structure and have acetate as a ligand. The chemical formula can be tentatively written as $\text{M}_x\text{O}(\text{CH}_3\text{COO})_y$;

CHECKPOINT

1 PTS

The particles formed by **M** have a similar metal-oxygen cluster structure

Considering the mass fraction of **M**, since **M** is hexacoordinate, if the oxygen cluster provides 1 coordination number, the ligand needs to provide 5 coordination numbers; if the structure of this particle is the same as $\text{Be}_4\text{O}(\text{NO}_3)_6$ and is a tetrahedral structure, and acetate is a bridging ligand, the coordination number of **M** is at most 4, so there are other ligands that provide coordination numbers.

Considering that the particles are prepared in water, consider that the ligand is water;

CHECKPOINT

1 PTS

There are other ligands that provide coordination numbers

CHECKPOINT

1 PTS

There is a ligand that is water

The chemical formula can be tentatively written as $M_xO(CH_3COO)_y(H_2O)_z$; considering that x is 4, at this time, acetate is a bridging ligand located on the edge of the tetrahedron, that is, $y=6$; in order to make the coordination number of **M** 6, each **M** needs to be additionally coordinated with two water molecules. Therefore, z is 8;

According to the mass fraction of 26.88%, the relative atomic mass of **M** is calculated to be 47.2, and there is no reasonable metal; therefore, the particle does not contain four **M**.

CHECKPOINT

1 PTS

Through mass fraction inference, the particle does not contain four **M**

Considering the decrease in symmetry, the particle contains three **M**. At this time, acetate acts as a bridging ligand, and two acetate groups can exist on the side of the planar triangle. Therefore, y can be 6 or 3, and the corresponding z that satisfies the coordination number can be 3 or 9.

According to the mass fraction, the relative atomic mass of **M** is calculated: when $y=6$, $z=3$, the atomic weight is 52.08, which is Cr; when $y=3$, $z=9$, it is 43.5, and there is no reasonable metal.

Therefore, the chemical formula of the particle is $Cr_3O(CH_3COO)_6(H_2O)_3$. Considering the common oxidation state of +3 for hexacoordinate Cr, the particle should carry a +1 charge. Therefore, options F and G are both incorrect.

CHECKPOINT

1 PTS

Considering the decrease in symmetry, the chemical formula of the particle is $Cr_3O(CH_3COO)_6(H_2O)_3$

CHECKPOINT

1 PTS

Considering the common oxidation state of +3 for hexacoordinate Cr, the particle should carry a +1 charge

The oxygen mass fraction in the particle is calculated to be 44.1%, option D is incorrect, and E is correct.